

Optimization of managed aquifer recharge (MAR) for drinking water supply in the Netherlands

Internship report



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Summary

The following report analyzes the use of Deltares' RTC-Tools to assess the improvements to a water supply system through the implementation of ASR wells. The document gives background on the project, its goals, and the methods of meeting these goals. Then, the study area and geology are characterized, providing the reader with context of the project. The optimization software is briefly explained, followed by a look into the model's application within the software. Finally, the results are produced and discussed, and the report concludes with next steps.

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1 Internship task

1.1 Project background

There are multiple stressors of water supply systems in the Netherlands, including rising demand, drought, and groundwater extraction limits [11]. To combat these issues and continue supplying the necessary quantity and quality of water to its customers, water suppliers must enhance their existing water infrastructure and explore new projects.

Managed Aquifer Recharge (MAR) is the concept of storing excess water underground for later use. By taking water that may otherwise evaporate or flow into the ocean and storing it in underlying aquifers, MAR can make water supply systems more resilient. One popular MAR method is the installation of Aquifer Storage and Recovery (ASR) wells, which inject excess water using wells and extract the water in times of need. Facing stressors on the water supply system, companies are exploring how ASR can bolster their water supply.

This report addresses the following question: How can Deltares' RTC-Tools software be used to assess the quantitative benefits of adding ASR wells to a water supply system?

1.2 Workflow

The first step in addressing this problem is to investigate the study area to roughly assess project demands and constraints. Next is to construct a conceptual model describing the physical system. After understanding the project demands and system functions, the next step is to build the elements of an RTC-Tools model: a Modelica .mo file, a goal table and input timeseries. Also important is assessing the model performance beyond graphical results. This required the implementation of some equations that detail the water supply reliability statistics so that comparisons between different model runs could be made. Finally, a thorough review of the results leads to conclusions on how the Strategic Heart model can improve water supply resiliency.

2 Study area and geology

2.1 Study area

The proposed study area is in the Province of Gelderland in eastern Holland, and, more specifically, just east of the capital city, Arnhem. The Province of Gelderland has a population of 2,173,425, and Arnhem, the second most populous municipality in Gelderland behind Nijmegen, has a population of 172,002 [6]. The Strategic Heart project is expected to supply water to nearby cities and meet a growing demand in the region: over 100,000 new homes are expected to be built by 2030 [11].

The project takes water from the river IJssel, which is an outflow of the river Rhine. The IJssel River has an average discharge of 250 m³/s [12]. The project may also take “additional water” from the Oude IJssel, a river that originates in Germany and meets the IJssel in Doesburg, a small city in between the two ASR zones. The river intake point is expected to be roughly 9 kilometers northeast of the start of the IJssel. From the intake point, the water will be transported to a nearby process basin. From there, the water will travel to a treatment plant, possibly in Ellecom, a town that sits on the western side of the IJssel. The western injection zone is proposed in the National Park Veluwezoom, complicating injection and extraction due to its status as a Natura 2000 area [11].

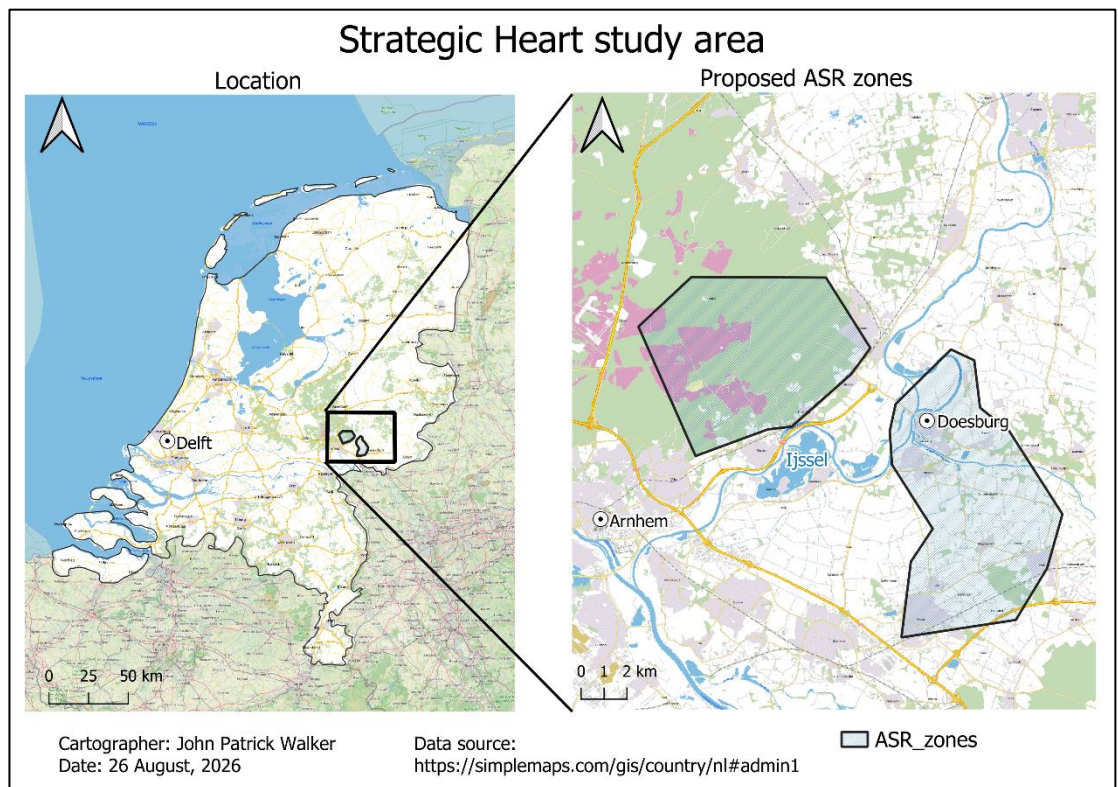


Figure 1 - Study area

2.2 Confining clay units

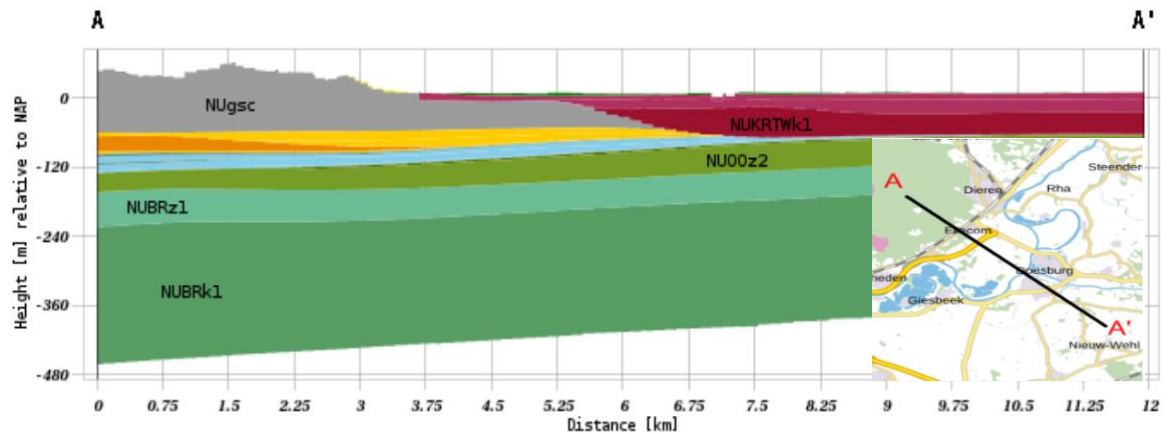


Figure 2 – Study area cross-section BRO REGIS II

The injection is planned to infiltrate a confined or semi-confined aquifer in each of the two ASR zones. Each zone is overlain by different confining layers. The above figure, collected from DINOLoket's BRO REGIS II model, demonstrates the two confining layers present beneath the ASR zones [1]. In maroon is the Twello Member of the Kreftenheye Formation and in yellow is the Waalre Formation.

East of the IJssel is the Twello Member of the Kreftenheye Formation, a clayey deposit formed in the middle to late Pleistocene. This member was deposited in a fluviodeltaic setting and has depths of roughly 60 – 75 meters. The Kreftenheye Formation has an average vertical hydraulic conductivity of 0.0005 – 0.001 m/d and an average thickness of 25m [4].

In the west is the third clayey unit of the Waalre Formation, another layer with clayey lithology from the late Pliocene to the early Pleistocene. This clay layer is found at depths of roughly 90 – 140 meters, has an average vertical hydraulic conductivity of 0.005 – 0.05 m/d, and an average thickness of 25 meters [5].

Well screens must exceed these depths in the two ASR zones to inject the treated water into the confined aquifer. The Twello member appears to be a good candidate for a confining aquitard considering its greater thickness and decreased conductivity, making it more resistant to leakage than the Waalre Formation. Further analysis must be done to assess injection rates into these two confined aquifers to avoid clay fracturing [6].

2.3 Aquifers

There are two principal aquifers that underlie the injection zones in the Strategic Heart project area. These can be seen in Figure 2 as the light blue formation and the uppermost green formation, the Maasluis Formation and Oosterhout Formation, respectively.

The Maasluis Formation is located west of the IJssel, underlying the Waalre clay unit. Its lithology is micaceous and calcareous very fine to fine sand [2]. According to DINOLoket, it has an average thickness of 35 meters, an average horizontal hydraulic conductivity of 5 – 10 m/d, and a transmissivity of 50 – 250 m²/d [2]. The Osterhout Formation, located east of the IJssel, underlies the Twello clay unit of the Krefentheye Formation. It is composed of very fine to medium sand, with an average thickness of 90 meters, an average horizontal conductivity of 5 – 10 m/d, and a transmissivity of 250 – 500 m/d [3]. The Osterhout Formation appears to be a good candidate as a storage aquifer because of its high transmissivity and its location directly beneath the Twello member aquitard.

3 Strategic Heart conceptual model and optimization methods

3.1 Conceptual model

The conceptual model has the following structure, as seen in Figure 3.

- Water intake from the river IJssel flows into a process basin before it is brought to a treatment plant;
- After treatment, water either flows directly to the distribution network, or, when supply exceeds demand, flows to the ASR wells for injection;
- Water can be extracted from the ASR storage and delivered to the distribution network in timesteps when demand exceeds the supply from the treatment plant.

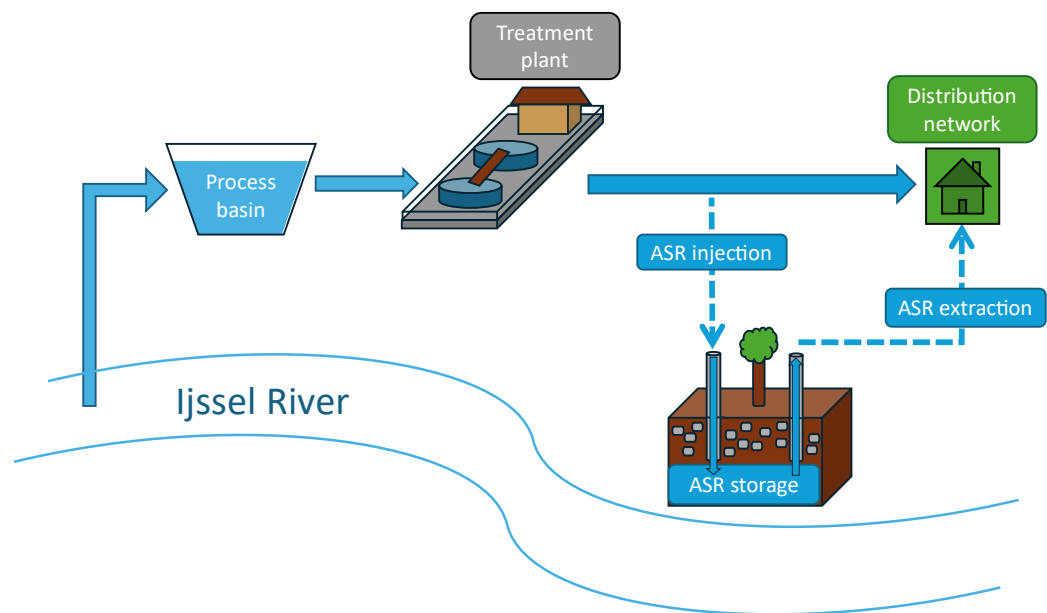


Figure 3 - Strategic Heart conceptual model

3.2 What is RTC-Tools?

RTC-Tools is a simulation and optimization software developed by Deltares. It is used to help water suppliers find optimal operating policies for their systems and to meet their goals. The RTC-Tools software used in this project is a part of their multi-objective optimization software, where the model equations are converted into constraints that are used by the solver. The solver solves each timestep sequentially, ensuring that the timestep it solves does not violate the goals/constraints of prior timesteps. The optimization problem uses linear constraints only, ensuring a globally optimal solution to the problem [5].

RTC-Tools uses an .mo file from Modelica Connection Editor that builds hydraulic connections between the system. Then, a goal table and timeseries input are built to apply constraints to the system based on the system's demands. Lastly, the system optimizer is run to solve the system using lexicographic goal programming, where the goals are ranked by priority and no goal can be satisfied if it violates a goal of higher priority. The solver then produces model results in the form of graphs and export files with results for each timestep.

RTC-Tools was incorporated into this project because it can take the model description and use optimization to see the outcome/results of the system. This approach informs the user of the model's performance and can be run multiple times with various inputs to see how the model responds to different inputs. RTC-Tools applies well to this scenario because it takes a complicated model and simplifies it to a water balance, finding the globally optimal solution and informing operating policy of the system.

3.3 Modelica model

This project applies RTC-Tools by first creating a modelica file that included a river intake boundary and an additional water boundary, each of which providing the input water for the system. The water intake is delivered to a process basin, followed by a treatment plant. The process basin has a minimum and maximum storage of one and three million cubic meters, respectively. The treatment plant has a maximum inflow of roughly $1.585 \text{ m}^3/\text{s}$, with 20% losses, meaning the maximum outflow is $1.265 \text{ m}^3/\text{s}$. The outflow from the treatment plant can be directed in two directions: either directly to the distribution network or to the ASR storage node. From the ASR storage, the treated water is extracted and delivered to the distribution network.

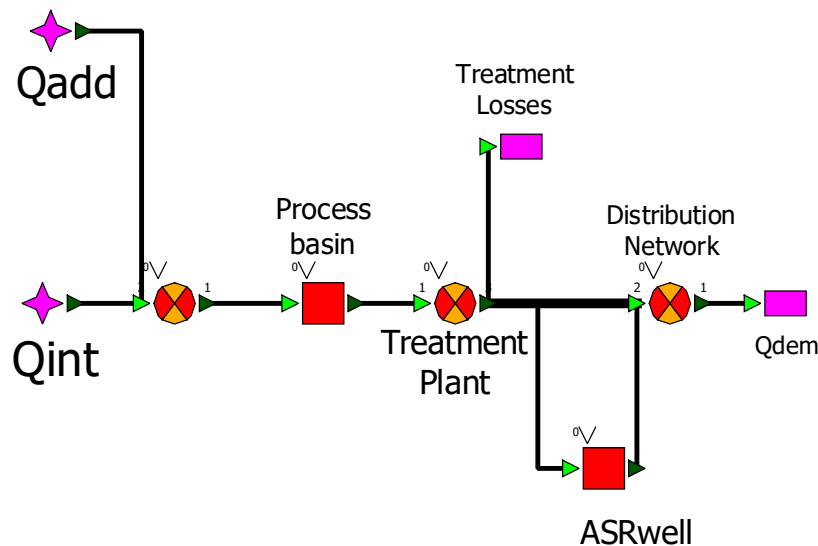


Figure 4 - Modelica model

3.4 Goal table

The goal table takes model equations and turns them into constraints by giving them goal types and target minimums and maximums. Then, lexicographic goal programming is used to solve the model by each priority, ensuring not to violate a higher priority goal. It can be seen in this instance that the highest priority goal is to ensure that the water sent to the distribution stays within reasonable bounds, especially not supplying more water than the demand.

The Strategic Heart goal table is structured as follows:

- Priorities 10-40 detail the system's operational capacities. This includes the treatment plant and river intake capacities, the storage volumes in the process basin and ASR aquifers, and the maximum injection and extraction rates for the ASR well

- The next highest priority is to minimize shortage, which is defined as the quantity demand at timestep i subtracted by the quantity distributed at timestep i . The ideal number is 0 for this goal, hence the minimization goal type.
- This goal is followed by system preferences based on client's needs, such as a minimal rate of change in the treatment plant, removal of unnecessary simultaneous ASR injection and extraction, and two other goals for the intake

The goal table can be altered to meet client needs, including additions with new goals, removing others, and adjusting the numbers used to determine these goals. The goal table gives the reader a thorough view of the solver's priorities when solving the problem at hand.

state	goal_type	target_min	target_max	priority
QDistribution	range	Qdistribution_min	Qdem	10
QTreatment	range		1.265	20
Qint	range		QintMax	20
Qadd	range		QaddMax	20
ASRVolume	range	ASRVolume_min	ASRVolume_max	30
ProcessBasinV	range	ProcessBasinMin	ProcessBasinMax	30
QASRInjection	range		InjectionMax	40
QASRExtracted	range		ExtractionMax	40
QShortage	minimization	0		50
QTreatment	rate_of_change	-0.01	0.01	60
QASRThroughput	minimization			70
Qadd	minimization			80
Qint	rate_of_change	-1	1	90

Table 1- Goal table (simplified for report)

3.5 Reliability statistics

While running the RTC-Tools software produces informative graphical outputs, additional statistics are needed to assess differences in system outputs beyond a visual comparison. These were made in the python script that blends the elements of the inputs with the RTC-Tools software to produce the outputs.

```
print("\nWater Supply Reliability")
print("-" * 55)
print(f"Total ASR water supplied : {total_asr_supplied / 1e6:8.3f} Mm³")
print(f"Maximum unmet demand      : {np.max(shortage):8.3f} m³/s")
print(f"Average unmet demand      : {np.mean(shortage):8.4f} m³/s")
print(f"Volumetric reliability      : {100-results['VolumeUnmetPct'][-1]:8.2f} %")
print(f"Percentage of weeks with >1% shortage : {results['TimeUnmetPct'][-1]:8.2f} %")
print(f"Years 1-5 reliability        : {first_5_reliability * 100:8.3f}%")
print(f"Years 6-10 reliability       : {second_5_reliability * 100:8.3f}%")
print(f"Summers with ASR supplied : {number_of_active_summers:8d}")
print(f"Average ASR per active summer : {average_asr_per_active_summer / 1e6:8.3f} Mm³/summer")
print(f"Additional people per active summer : {additional_people_per_summer:8.0f} people")
print(f"Total water supplied         : {total_water_supplied / 1e6:8.3f} Mm³")
print(f"Percentage supplied by ASR: {asr_percentage:8.2f} %")
```

Equation 1 - Reliability statistics python equations

First, it was important to quantify the additional water produced by the ASR system. Then, a calculation of the maximum and average unmet demand were implemented to assess the shortcomings of the system. Next, two types of reliability were considered: volumetric and temporal. The volumetric reliability simply divided the quantity distributed by the quantity

demand to show the percentage of demand being met by the system. The temporal reliability counts the weeks with shortage, or where distribution was short of demand by more than 1%, and divides that by the total number of weeks. These two calculations paint a clearer picture of the improvements to the system after implementing ASR, especially when assessing the reliability over the final five years of the system. Finally, a calculation was implemented to put the improvements to the system into perspective by quantifying the additional number of people served by the ASR system. This was done by dividing the amount of water per ASR cycle by the length of the ASR cycle in days times the per person demand of .125 m³ per day, seen in the equation below.

$$\text{Additional people supplied by ASR} = \frac{\text{average ASR extraction per cycle}}{(\text{days of ASR per cycle} * \text{per person demand})}$$

Equation 2 - Additional people supplied by ASR

The equations above aim to bridge the gap between complex graphical output of an optimization software and quick assessment of the water supply system performance to allow for easier project evaluation by those unfamiliar with the optimization software.

4 Model results

4.1 Results introduction

The following results scenarios are two simulations of the model using RTC-Tools. The input timeseries used in each scenario is designed by the author with the intent of demonstrating ASR's capabilities to improve water supply. The end user can adjust the model to their needs, including the following input values:

- injection/extraction limits
- ASR and process basin storage capacities
- Demand and minimum distribution rates
- treatment limits

In these scenarios, the demand is a function of the treatment plant capabilities, with average demand roughly proportional to the treatment capacity and a ten percent increase in demand during summer months, paired with a ten percent decrease in demand during winter months. The demand timeseries is accompanied by a minimum distribution threshold, which is 90% of the demand in the relevant timestep. The ASR extraction can only occur after one year of injection, which creates a freshwater bubble within the aquifer.

The results of importance in the following results are the ASR injection and extraction rates, the treatment plant rate of change, and the comparison between water supply reliability in scenarios with and without ASR.

4.2 Constant demand scenario

The following scenario shows a water supply network across ten years with constant demand. The first figure (Figure 5 - Constant demand NO ASR results) shows the ten-year simulation of the water supply network without ASR, accompanied by the statistics of the system (Table 4 - Rising demand NO ASR statistics). The second figure shows the simulation of the water supply network with ASR supplementation. The demand curve in this scenario has a roughly 10% increases in summer and roughly 10% decrease in winter.

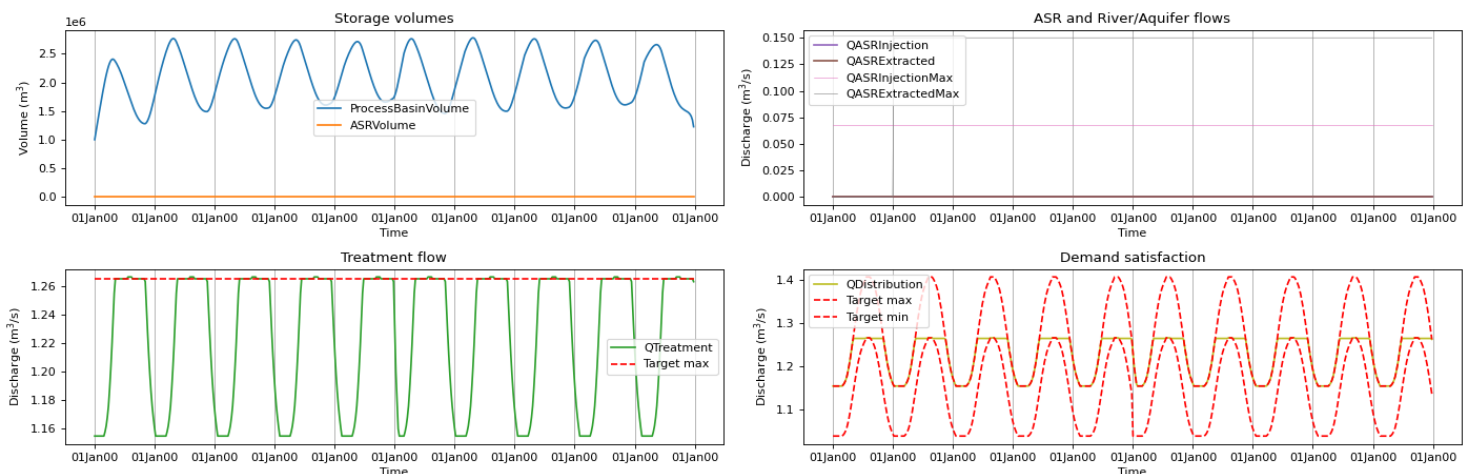


Figure 5 - Constant demand NO ASR results

The figure above depicts the results of the scenario being run with ASR inactive. The upper right graph shows zero ASR water injected nor supplied. The bottom right graph shows how

QDistribution compares with the demand curve (upper dashed red line). QDistribution clearly falls well short of the desired delivery, barely meeting 90% of demand (bottom dashed red line) in peak summer demand. There is also a 100 lps rate of change in treatment, which is an undesirable outcome. The following water supply reliability statistics quantify the system results.

Water Supply Reliability

Maximum unmet demand	:	0.141 m ³ /s
Average unmet demand	:	0.0477 m ³ /s
Total ASR water supplied	:	-0.000 Mm ³
Worst weekly shortage	:	10.00 %
Maximum oversupply	:	-1.845e-09 m ³ /s
Volumetric reliability	:	96.25 %
Volumetric unmet demand	:	3.75 %
Percentage of weeks with >1% shortage	:	250 / 522
Temporal unmet demand (>1%)	:	47.89 %
Years 1-5 reliability	:	96.253%
Years 6-10 reliability	:	96.254%
Full 10-year reliability	:	96.253%

Table 2 - Constant demand NO ASR statistics

The statistics show a reliability of 96.25%, meaning the system delivers 96.25% of its demand across ten years. It has shortages of >1% of demand in 47.89% of weeks, meaning nearly half of the ten-year period fails to deliver full demand. This result is then compared with the same scenario in the figure below, though with ASR activated.

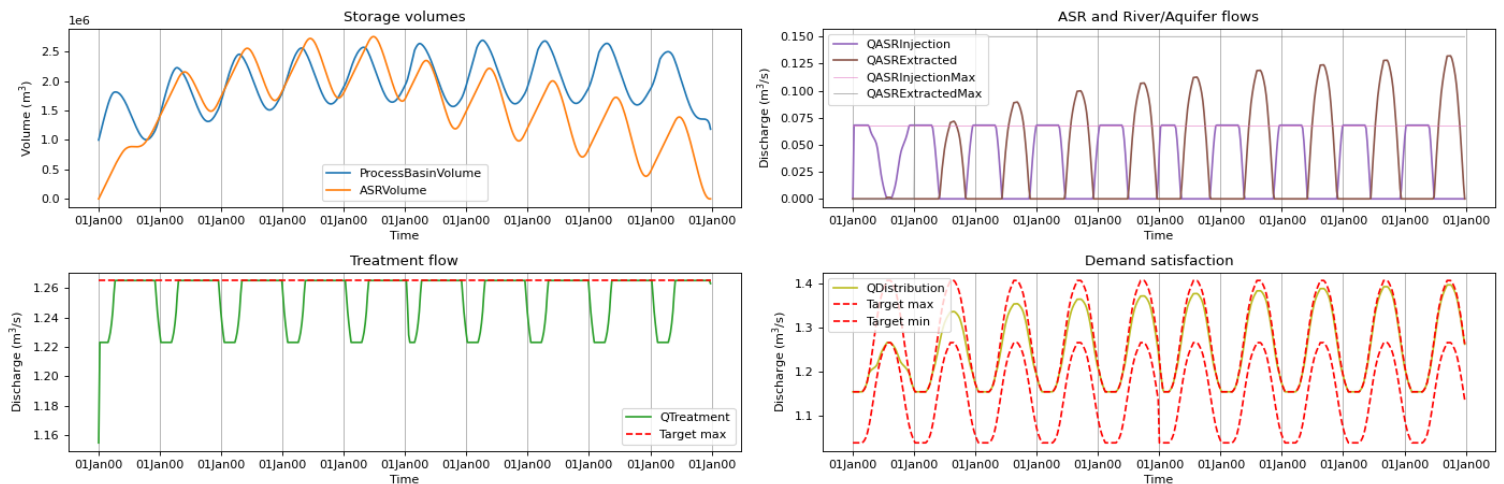


Figure 6 - Constant demand ASR results

The above results show a QDistribution curve that comes much closer to meeting the ideal demand curve. Similarly, the treatment rate of change is at a much more desirable 45 lps. These changes are due to ASR injection and extraction, which can be seen in the upper right graph. The ASR appears to follow a seasonal pattern, injecting in the months of low demand (December – March) and extracting when demand exceeds supply (June – August). The following statistics can demonstrate the improvements to the system.

Water Supply Reliability

Total ASR water supplied :	10.087 Mm ³
Maximum unmet demand :	0.141 m ³ /s
Average unmet demand :	0.0189 m ³ /s
Volumetric reliability :	98.52 %
Percentage of weeks with >1% shortage :	41.57 %
Years 1-5 reliability :	97.677%
Years 6-10 reliability :	99.353%
Summers with ASR supplied :	10
Average ASR per active summer :	1.009 Mm ³ /summer
Additional people per active summer :	89664 people
Total water supplied :	395.638 Mm ³
Percentage supplied by ASR:	2.55 %

Table 3 - Constant demand ASR statistics

The table above shows the results of the constant demand scenario using ASR. Activating ASR leads to an additional supply of more than 10 million m³. This raises the reliability of the system by more than 2%. This also reduces the percentage of weeks with >1% shortage by 6%, meaning the water supply system becomes more reliable in time and in volume.

4.3 Rising demand scenario

The following scenario shows a water supply network across ten years with rising demand. As above, the first figure, Figure 7, shows the ten-year simulation of the water supply network without ASR, accompanied by the statistics of the system. The second figure, Figure 8, shows the simulation of the water supply network with ASR supplementation. The demand curve in this scenario has a roughly 5% increase in summer and roughly 5% decrease in winter, as opposed to ten percent in the constant demand scenario above.

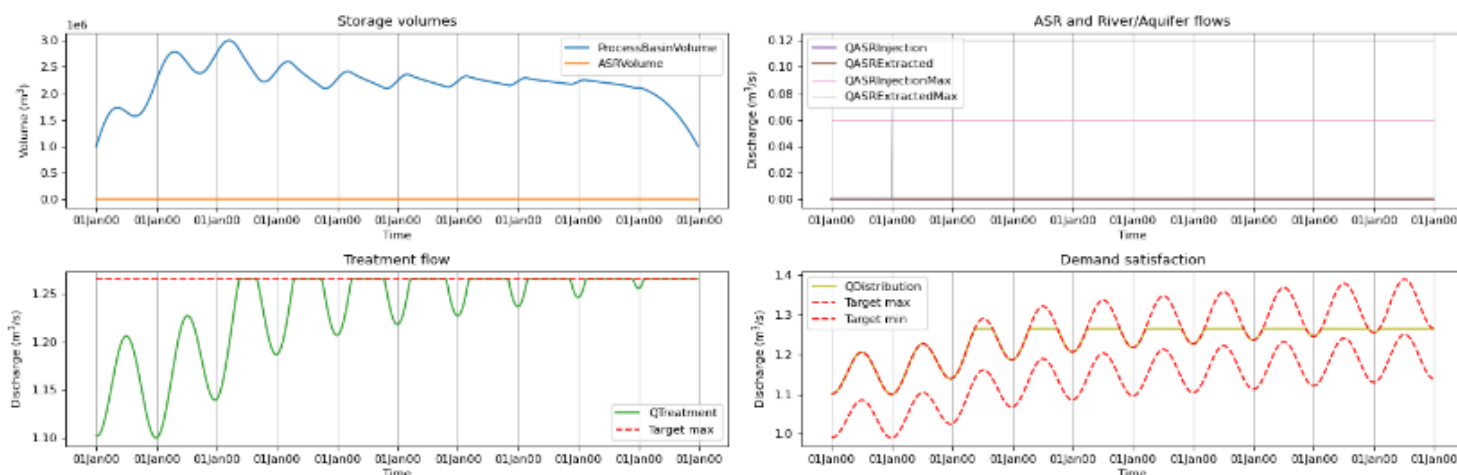


Figure 7 - Rising demand NO ASR results

Figure 7 shows the results of the rising demand NO ASR simulation across ten years. Note the flat lines of distributed water that continuously fall shorter of meeting demand requirements as time goes on. This trend of supply shortage, coupled with oscillating treatment flow, shows a water supply system that is not performing at optimal capacity, and this is backed up by the statistics below.

Water Supply Reliability

Total ASR water supplied :	-0.000 Mm ³
Maximum unmet demand :	0.125 m ³ /s
Average unmet demand :	0.0273 m ³ /s
Volumetric reliability :	97.84 %
Percentage of weeks with >1% shortage :	42.91 %
Years 1-5 reliability :	99.223%
Years 6-10 reliability :	96.544%
Summers with ASR supplied :	2
Average ASR per active summer :	-0.000 Mm ³ /summer
Additional people per active summer :	-0 people
Total water supplied :	389.496 Mm ³
Percentage supplied by ASR:	-0.00 %

Table 4 - Rising demand NO ASR statistics

The water supply reliability statistics show 42.91% of weeks with a >1% shortage, highlighting the vulnerability of the system. Overall, the system supplies 97.84% of its expected demand. The reliability decreases from years 1-5 to years 6-10 due to rising demand, representative of the increasing demand within the Strategic Heart study area. This simulation is compared with the ASR scenario below.

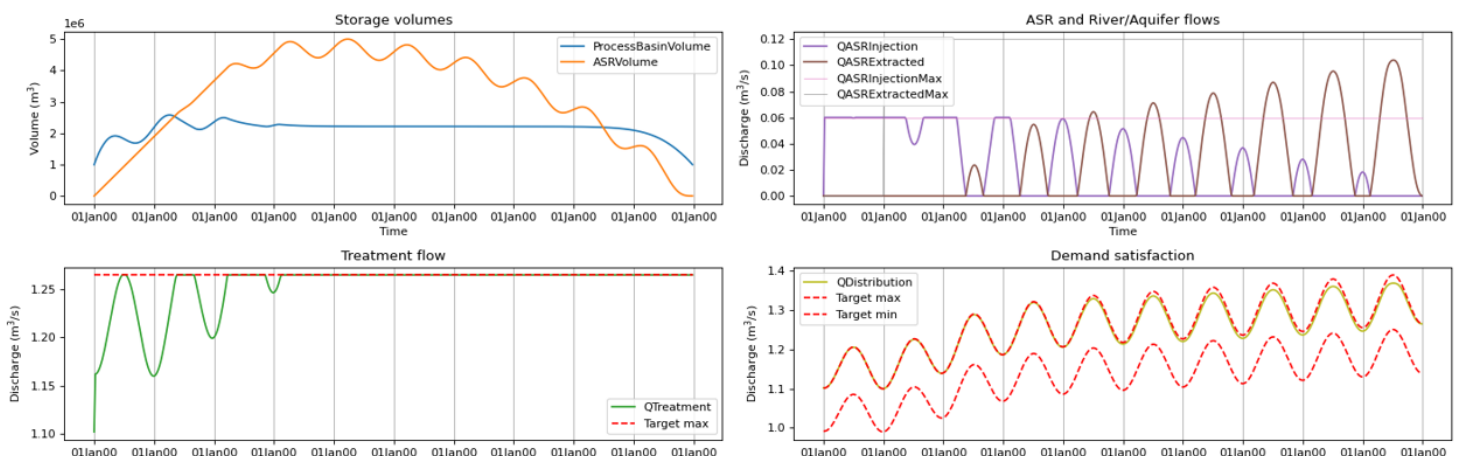


Figure 8 - Rising demand ASR results

Figure 8 demonstrates the water supply system with a greatly improved supply reliability. The injection occurs in winter months when treatment exceeds demand, and extraction occurs when demand exceeds treatment. QDistribution line is much closer to the demand curve, coming far closer to meeting full demand in the summer months. The statistics below further explain the graphs.

Water Supply Reliability

Total ASR water supplied :	7.101 Mm ³
Maximum unmet demand :	0.021 m ³ /s
Average unmet demand :	0.0063 m ³ /s
Volumetric reliability :	99.50 %
Percentage of weeks with >1% shortage :	10.15 %
Years 1-5 reliability :	99.862%
Years 6-10 reliability :	99.156%
Summers with ASR supplied :	8
Average ASR per active summer :	0.888 Mm ³ /summer
Additional people per active summer :	78896 people
Total water supplied :	396.104 Mm ³
Percentage supplied by ASR:	1.79 %

Table 5 - Rising demand ASR statistics

The supply statistics back up the improved results graphs above. The system adds more than 7 million m³ of water supply over ten years. The percentage of weeks with shortage decrease by over 30%. The volumetric reliability increases by nearly 2%, and in years 6-10 an increase in reliability of more than 2.5%, showing how this system is equipped to meet rising demand. This scenario is estimated to provide water for an additional 78,896 people during summer months than the scenario without ASR. These statistics confirm that the water supply can improve reliability by adding ASR wells.

5 Conclusion

5.1 Discussion

The ASR scenarios each lead to a decreased rate of change of the treatment plant, meeting one of the client's goals. While ASR does lead to additional water supply, the reliability is typically limited by the injection limits and the treatment plant capacity. This points to areas where the client can investigate changes to increase water supply from ASR.

The treatment plant still provided most of the water to the system (>97%) in all scenarios, meaning the ASR system only makes marginal gains. However, these gains, when paired with other supply resiliency methods like further ASR/groundwater exploration, water conservation methods, and surface water capture, can be the gains that provide for a system with rising demand in a changing climate.

Using Deltares' RTC-Tools to assess the benefits of ASR to a water supply system allows for quick graphical visualization and comparisons. There were challenges in implementation of the system in Modelica, ensuring that the flow of water matched the intentions of the water supplier, and that the goals were being met by the system. Once the system was operating as intended, the next difficulty was understanding how to compare different scenarios. This was solved by implementing the supply reliability statistics diagnostic within the python script, allowing for quick diagnosis of model performance.

5.2 Next steps

The results presented in this report provide the basis for exploration of the benefits of ASR well implementation to a water supply system. The immediate next steps are to work with the client to better understand and implement their projections for demand and their river intake, process basin, and treatment plant capacities. Likewise, a more thorough hydrogeological assessment must be done to improve estimations for injection and extraction capacity, allowing a more accurate estimation of ASR capabilities. From there, an analysis of the costs of well implementation and the gains of added water supply provide the tools needed for the client to make an informed decision on the project.

5.3 How this project can be continued

The Strategic Heart model files are located in the following github repository: [walker23jack-collab\JPW_rtc_git](https://github.com/walker23jack-collab/JPW_rtc_git). These files include the goal table, the input timeseries for both demand scenarios, the plot table, the .mo modelica file, and the python script, as well as a README with instructions for use. The project must be run with file names specific to the python script, though the scripts can be edited to include more information or alter system functions.

There is a weekly model, from which the results are presented above, and an hourly model. To run the hourly model, the following must be done:

1. The name of the timeseries import file must be changed from "timeseries_import_hourly" to "timeseries_import."
2. The "runRTC-Tools-WinPython" batch command must be opened and edited to run the "StrategicHeartHourly.py" python script instead of the currently-used "StrategicHeartWeekly.py"
3. Further documentation can be found here: <https://rtc-tools.readthedocs.io/en/stable/>

The continuation of this project involves three crucial aspects. First, communication with the client to assess real demand projections for the next ten years. This can inform the timeseries input file used to run the model to give more realistic results for the system's water supply needs. Secondly, a thorough hydrogeological assessment of the underlying clay layers and aquifer system will give better limits for injection and extraction. Considering that ASR injection is the model constraint that most severely limits the increase in water supply from the ASR system, it is greatly important to determine an accurate injection capacity. Last is the implementation of a MODFLOW model using the presented hydrogeological units to validate the data produced by the model and induce a further reliability into the projections.

Future projects can also include an analysis of hourly timesteps using the model included in the github, temperature models, groundwater flow models, and water contamination impacts on river intake.

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